



Event Frequency Layer

Definition Document and Project Plan — MKM-EF-001

MKM Research Labs

Version 2.1 — 22-July-2026

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Contents

Status	4
1 What changed from version 1	5
1.1 What version 2.1 changes	5
1.2 What version 2.0 changed from version 1	5
2 Background and problem statement	6
2.1 Current state	6
2.2 The defect, with evidence	6
2.3 What is already correct	7
2.4 The pattern already exists in this repository	7
2.5 Remedy	8
3 Objectives	8
4 Scope	8
4.1 Locked decisions	8
4.2 In scope	10
4.3 Chain position	10
4.4 Out of scope for this phase	10
5 Functional specification	11
5.1 Package layout	11
5.2 Configuration	11
5.3 Event-definition alignment	11
5.4 Frequency abstraction	12
5.5 Calibration pipeline	12
5.6 Persistence	13
5.7 The event catalogue	13
5.8 The Monte Carlo year simulation	13
5.9 The closed form, and why it is a test rather than the answer	14
5.10 Wiring	14
5.11 Extension points	15
6 The λ provenance problem	15
7 Measured results	16
7.1 Cost and accuracy of the year simulation	16
7.2 Repricing: withdrawn, not yet measurable	16
7.3 Extraction against the existing calculation	17
8 Non-functional requirements	17

9 Project plan	18
9.1 Stage acceptance	19
10 Migration landmines	19
11 Governance and registration	21
12 Risks and assumptions	21
13 Glossary	22

Status

Document type	Definition Document and Project Plan
Component	Event Frequency Model — MKM-EF-001 (new)
Version	2.1 — supersedes v1.0 of 22 July 2026
Status	Stage 1 implemented; plan reconciled with the build
Owner	CSO, MKM Research Labs
Proposed tier	1 (Maximum)

This document proposes a new model for the inventory and a staged plan to remove a pricing defect in the existing flood chain. Stage 1 is built and under test; Section 7 reports what it measures. Nothing yet feeds a price.

1 What changed from version 1

1.1 What version 2.1 changes

Version 2.0 was written before any code existed. Building it moved four things, one of which was an outright error in version 2.0's design.

#	Change	Driver
C7	λ is a property of the catchment, not of a gauge. Version 2.0 had per-gauge peaks-over-threshold calibration feeding λ into pricing. That is wrong: a per-gauge exceedance rate is <i>already</i> $\lambda \times P(\text{exceed} \mid \text{event})$, so using it as λ squares the conditional and double-counts. λ is now a seeded per-catchment rate, and per-gauge extraction became the validation arm	A storm arrives over the catchment and reaches every gauge in it. What differs per gauge is the conditional response, not the arrival rate
C8	The Monte Carlo year simulation is the engine , and has moved forward out of Track B. Version 2.0 deferred the sampler and priced off the closed form; the order is now reversed	The closed form is the simulation's <i>exact expectation</i> , not an approximation of it, so it is worth more as a check than as an answer. The simulation additionally yields the annual distribution, which the closed form cannot
C9	Ten thousand simulated years , and the reconciliation gate is expressed in sampling standard errors rather than as a fixed percentage	Measured on the target hardware. A fixed 2% band false-alarmed on 17% of runs at ten thousand years while never binding at a million
C10	Landmine L3 resolved: <code>num_storms</code> is kept and <code>num_events</code> added beside it	The storm/event distinction becomes visible in the data rather than hidden in a redefinition of a field existing consumers already read

1.2 What version 2.0 changed from version 1

Version 1 diagnosed the problem correctly. Version 2 keeps the diagnosis and reworks the plan against what the codebase actually does. Six substantive changes:

#	Change	Driver
C1	The layer is no longer greenfield — it is a port of the existing seismic and fire occurrence pattern onto the storm chain	<code>src/models/seismic/occurrence/</code> and <code>src/models/fire/initiation.py</code> already implement precisely the proposed $\lambda \rightarrow \text{Poisson} \rightarrow \text{event} \rightarrow \text{severity}$ construction. Storm and flood are the outlier, not the pioneer.

#	Change	Driver
C2	Delivery splits into Track A (analytic annualisation) and Track B (ELT → YLT); only Track A is in scope	Version 1 went straight to the YLT sampler before any pricing benefit landed. Track A fixes pricing at one node without touching port generation.
C3	Scope is a generic per-peril frequency layer , calibrated for storm first	The wind leg carries the identical defect; fire and seismic already carry their own λ . A flood-only layer would need a second frequency layer within months.
C4	New Section 6: λ provenance is currently circular on synthetic catchments	The “historical” gauge series is generated <i>from</i> the frequency field it would be used to calibrate. This materially changes what Stage 1 can claim.
C5	Version 1’s “critical alignment task” is already broken in a specific, locatable way — sequences are shredded into storms before the denominator is taken	<code>src/models/hazard/io/_load.py:52</code>
C6	Non-functional requirements restated against the actual house rules and audit tooling	Version 1 cited <code>split_audit.py</code> , which no longer exists; specified 100% branch coverage against a house rule of $\geq 99\%$ line; and did not mention the database seam at all.

2 Background and problem statement

2.1 Current state

The simulator generates synthetic storm events and evaluates, at each river gauge, whether a storm produces a flood. Insurance cost is priced as the ratio of flooding storms to total storms. This is a **conditional probability** — $P(\text{flood} \mid \text{storm})$ — and is dimensionless with respect to time.

Two gauges with identical conditional probability therefore receive identical prices even if one experiences eight qualifying storms a year and the other one. The true annualised risk differs by a factor of eight.

2.2 The defect, with evidence

The defect is not conceptual. It is written out in three places.

Location	Code	Meaning
src/port/src/property/hc/ pricing/_process.py:54	spread_bps = (flood_count / num_storms) * 10000	The property PRS spread is $P(\text{flood} \mid \text{storm})$, expressed in basis points
src/port/src/property/hc/ pricing/_process.py:250	return_period_yrs = num_ storms / flood_count	The assumption stated explicitly: one storm = one year , $\lambda \equiv 1.0$
src/models/hazard/ builder.py:109--141	GEV fitted to the peak level of <i>every</i> simulated storm; the resulting exceedance probability assigned to annual_exceedance_prob and annual_hazard_rate*	A per-event conditional is re-labelled <i>annual</i> at the point of assignment

The same ratio construction appears on the gauge basis leg (`_process.py:119`) and the wind leg (`_process.py:70`, via `_wind_union`).

2.3 What is already correct

The downstream plumbing is rate-shaped and needs no rework:

- `src/models/hazard/gev.py:66` — `compute_term_structure` is already a Poisson compounding, $P(\geq 1 \text{ flood by year } t) = 1 - e^{-\lambda t}$.
- `src/models/hazard/prs_analytical.py:100` — already converts an annual probability to a continuous hazard rate via $-\log(1 - p)$.

Both are being fed a per-event number. **The remedy is a substitution at one node, not a pricer rewrite.**

2.4 The pattern already exists in this repository

Two production models already do what version 1 proposed to invent:

- `src/models/seismic/occurrence/` — `_rates.py` computes a per-asset annual λ ; `_simulate.py:89` draws `rng.poisson(lambda_annual * horizon_years, size=n_sim)` and instantiates events where the count is at least one; severity and damage follow.
- `src/models/fire/initiation.py` — the same construction, labelled “Model A — Poisson initiation”.

MKM-EF-001 should mirror this package shape rather than introduce a parallel abstraction with different conventions.

2.5 Remedy

$$\text{Annual flood frequency}_g = \lambda_g \times P(\text{flood} \mid \text{event}, g) \quad (1)$$

$$\text{Annual exceedance probability}_g = 1 - \exp(-\lambda_g P(\text{flood} \mid \text{event}, g)) \quad (2)$$

$$\text{Pure premium}_g = \lambda_g \times P(\text{flood} \mid \text{event}, g) \times \mathbb{E}[\text{loss} \mid \text{flood}, g] \quad (3)$$

with λ_g the calibrated annual **event** arrival rate — where “event” means the insurance hours-clause event, not the individual storm (Section 5.3).

3 Objectives

#	Objective	Success measure
O1	Price reflects event frequency	AAL differs proportionally between high- and low-frequency gauges with equal conditional flood probability
O2	Honest return periods	<code>return_period_years</code> derived from λ and the conditional, not from the storm count; the 100-year clamp reviewed
O3	Independently validatable components	Frequency backtestable separately from vulnerability; MKM-GH-001 outputs renamed so nothing claims to be annual that is not
O4	Compliance by construction	λ_g provenance recorded and reproducible (BCBS 239), <i>including honest recording of generator-derived provenance</i> ; validation artefacts auto-generated (SR 11-7)
O5	No new technical debt	House rules R1–R6 satisfied; calibration persisted through the <code>database</code> seam, not JSON
O6	Quantified repricing before switchover	Per-gauge and per-property parallel-run delta report produced <i>before</i> the legacy metric is retired

4 Scope

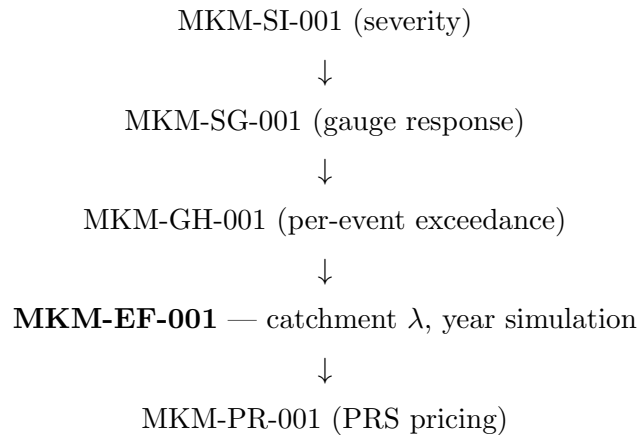
4.1 Locked decisions

Decision	Choice	Rationale
Pricing engine	Monte Carlo year simulation. Draw how many qualifying events arrive in a year, ask of each whether it floods, repeat. The annual flood probability is the fraction of simulated years that flooded	Retains the platform’s simulation framework rather than replacing it with a formula, and yields the whole annual distribution — occurrence and aggregate views, and return periods — not just a mean
Closed form	Reconciliation self-test, not an alternative answer	It is the simulation’s exact expectation. Any disagreement beyond sampling error means one of the two is wrong, which makes it a real check rather than a restatement
λ granularity	Per catchment , not per gauge	A per-gauge exceedance rate is already $\lambda \times P(\text{exceed} \mid \text{event})$; using it as λ would square the conditional
λ source	Configured per-catchment seed (Thames 4.5 events per year), with per-gauge extraction as the validation arm	Per house convention for a new probabilistic model. The data-driven estimator is blocked by Section 6 on synthetic catchments in any case
Peril scope	Generic abstraction with per-peril λ ; calibrate storm first	Wind carries the same defect; fire and seismic already have λ and fold in later
Event aggregation	Maximum level across the storms of a sequence	A PRS pays on a level being breached, and a week containing two breaches is one breach of the contract, not two
Denominator	Keep num_storms, add num_events	Existing consumers read num_storms; redefining it in place would change their meaning silently
Chain position	Downstream of storm severity	Nothing upstream changes; one switch point
Model ID	MKM-EF-001 — <i>Event Frequency Model</i>	EF is unused in the model inventory
Tier	1 (proposed)	λ is a linear multiplier on every PRS spread — more material than MKM-SI-001 (tier 2, “not directly into pricing”). Peers at tier 1: MKM-GH-001, MKM-PR-001, MKM-DE-001

4.2 In scope

- Frequency abstraction with Poisson and Negative Binomial implementations, per peril.
- Calibration pipeline: peaks-over-threshold extraction, declustering, arrival-rate estimation, dispersion diagnostics, per-gauge model selection.
- **Event-definition alignment**: the hours-clause event (a storm *sequence*), not the individual storm, becomes the unit of both λ and the conditional.
- The annualisation seam and its wiring into MKM-GH-001.
- Repricing of the property and gauge PRS legs.
- Calibration persistence and provenance through the **database** seam.
- Validation module: dispersion test, count backtest, threshold sensitivity, reconciliation, POT round-trip.
- Configuration schema extension in a new **config/frequency/** package.
- Governance pack: inventory entry, model documentation, registry wiring.

4.3 Chain position



Physically, frequency is a *sibling* of severity rather than a descendant. As an implementation seam, downstream is the right choice: nothing upstream changes, MKM-GH-001 is merely **relabelled** — its outputs become honest per-event conditionals — and **builder.py** is the single switch point.

Inventory **upstream_models** and **downstream_models** must be updated on MKM-GH-001 and MKM-PR-001 accordingly; chain consistency is scanned by full-audit section 4.7.

4.4 Out of scope for this phase

- **Track B**: ELT export, YLT sampler, AEP construction, portfolio aggregation, cross-gauge count correlation.
- Non-stationary or climate-conditioned $\lambda(t)$. Architected for, not implemented.
- Seasonal and non-homogeneous Poisson rates.
- Wind, fire and seismic λ *calibration* — the abstraction accommodates them; only storm is calibrated in this phase.
- Changes to the inundation model, the spatial model or the DEM integration.
- Multi-year contract pricing and discounting.

5 Functional specification

5.1 Package layout

Mirrors `src/models/seismic/occurrence/`. Every file under 300 lines (R2); no callable code in any `__init__.py` (R4); canonical copyright header on every file (R5).

```

1 src/models/frequency/
2   __init__.py           # module docstring + re-exports only
3   datastructures/
4     _rate.py            # FittedRate: lambda, family, params, provenance
5     _diagnostics.py     # dispersion index, log-likelihood, AIC
6   pot/
7     _threshold.py       # threshold selection + stability diagnostics
8     _decluster.py       # minimum inter-event separation
9     _extract.py         # series -> declustered exceedances -> annual counts
10  families/
11    _poisson.py
12    _negbin.py
13    _select.py          # dispersion test + AIC tiebreak; logs any override
14  annualise.py          # p_annual = 1 - exp(-lambda * p_event)  <-- the seam
15  calibrate.py          # pure function of (series, config) -> FittedRate

```

Listing 1: Proposed package layout

`annualise.py` is deliberately tiny and deliberately the only place the composition happens. Frequency logic must not leak into the event simulator or the pricing modules.

5.2 Configuration

A new `config/frequency/` package with `_schema.py` and `_loader.py`, following the hydrated-dataclass pattern of `config/seismic/`. It owns, per house rule R1:

- Target mean exceedance rate band for threshold selection.
- Declustering window (minimum inter-event separation).
- Dispersion-test critical value; AIC tiebreak policy.
- Minimum record length for per-gauge calibration; regional fallback λ .
- Return-period grid for output tables.
- Per-peril family overrides, with the override itself logged (SR 11-7).

It also absorbs an existing orphan: `src/models/intensity/distribution/_core.py:198` carries `storms_per_year: float = 20` as a default argument — an undocumented frequency assumption living inside the severity model, and an R1 breach. That parameter moves here and the severity model takes it as an injected value.

5.3 Event-definition alignment

This is not a hypothetical risk. It is already wrong, and locatably so.

MKM-SS-001 produces **sequences** — one to five related storms fitted inside a 168-hour insurance hours clause (`EVENT_WINDOW_HOURS` in `config/port/_storm.py`). But `src/models/hazard/io/_load.py:52--61` flattens every sequence into its member storms before the hazard model sees them.

`num_storms` is therefore one to five times the event count, and today's denominator is **neither events nor years**.

Built. The remedy as delivered:

1. `load_storms_from_sequences` tags each storm with its `sequence_id`. The key is **added, not substituted**, so every existing consumer of the storm fields is untouched. An untagged sequence falls back to a positional identity, so two untagged sequences cannot silently merge into one event.
2. `event_id` and `count_events` live with the loader that writes the tag, and the frequency aggregator imports them, so the fallback rule cannot drift between the two. This also keeps the dependency pointing downhill: frequency imports hazard, never the reverse.
3. The conditional is evaluated per event — the **maximum** peak level across the member storms, not one row per storm.
4. Grouping is by storm **identity**, not position. The two coincide today, but position would break silently if response ordering ever changed.
5. λ is quoted in **hours-clause events per year**, matching that unit exactly.
6. `num_storms` is kept and `num_events` added beside it in both the hazard metadata and the build summary.

The physics is untouched: each storm is still routed through the gauge response model individually. Only the grouping changes.

This step reprices on its own, before λ is applied, because the denominator changes. It was sequenced first precisely so its effect could be measured apart from the λ effect — and doing so was worthwhile, because the two turned out to push the *same* way rather than offsetting (Section 7).

5.4 Frequency abstraction

A single interface, per peril, following the seismic rates/simulate split:

- `fit(counts: AnnualCountSeries) -> FittedRate`
- `annual_rate() -> float`
- `sample_annual_count(rng) -> int` (*unused in Track A; required by Track B*)
- `diagnostics() -> FrequencyDiagnostics`

Implementation	Distribution	Selected when
PoissonFrequency	Poisson(λ)	Dispersion index ≈ 1 (default)
NegativeBinomialFrequency	NegBin(r, p)	Overdispersed counts (variance $>$ mean)

Selection is data-driven per gauge via the dispersion index with an AIC tiebreak, recorded in the calibration output. A configuration override may force a family; the override is logged with its justification.

5.5 Calibration pipeline

Per gauge, from the gauge level record:

1. **Threshold selection** — the threshold yielding a target mean exceedance rate, anchored on bankfull discharge where available. Diagnostics: mean-excess and threshold-stability.

2. **Declustering** — enforce a minimum inter-event separation so peaks are independent. Record the rule applied.
3. **Arrival rate** — $\hat{\lambda} = \text{exceedances} / \text{years of record}$; the annual count series is retained for dispersion testing.
4. **Dispersion test and family selection.**
5. **Severity cross-check (validation only)** — fit a GPD to exceedance magnitudes and compare against the simulator’s implied severity at that gauge. Divergence beyond tolerance raises a validation flag; it never silently recalibrates anything.

Prior art to fix, not to write from scratch. `src/models/statistics/timeseries.py:103-122` already computes a per-gauge `frequency_per_year` for alert, warning and severe. It counts **days over threshold**, not declustered events — peaks-over-threshold with step 2 missing — and therefore overstates λ by roughly the mean event duration in days. Stage 1 is a fix-and-promote of this calculation into the new package, with the existing consumers repointed.

Provenance (BCBS 239). Every fitted rate carries: source dataset identifier and version, record period, threshold, declustering rule, family selected and why, fit timestamp, code version, configuration hash, and a **provenance class** — `observed`, `generator-derived` or `regional-fallback`. Calibration is a pure function of (data, configuration): re-running with identical inputs must reproduce identical outputs, regression-tested against a golden master.

5.6 Persistence

Fitted rates and their provenance are written and read **exclusively through the `src/database/seam` (R6)**. No JSON artefact is created at any point — full-audit section 4.5 is walking JSON writes toward zero tolerance, and a new model must be born database-native.

Deliverables: a new table plus Alembic migration following the existing gauge hazard-curve migration, with matching entries in the ORM model and column maps, and accessor functions in the `database` package API.

5.7 The event catalogue

One row per event, one column per gauge, holding the highest level that event drove at that gauge. This is what the year simulation resamples from.

Because every gauge’s outcome for a given event sits in the **same row**, drawing an event row draws a spatially coherent storm. The correlation between gauges is carried by the catalogue itself and does not have to be modelled again downstream.

The catalogue also exposes the conditional half of the decomposition directly: $P(\text{flood at } g \mid \text{event})$ is the fraction of catalogue rows whose level at g clears the trigger.

5.8 The Monte Carlo year simulation

For each simulated year: draw $N \sim \text{Poisson}(\lambda)$ qualifying events, resample N catalogue rows with replacement, and record how many of them flood. The annual flood probability is the fraction of years with at least one flood.

The draw is separated from its application. `draw_event_years` runs once per run; `apply_catalogue` runs per subject against those same draws. A storm arrives over the catchment and reaches every gauge and property in it, so one set of draws must serve every subject. Drawing independently per subject would discard the spatial correlation the catalogue supplies: measured, two gauges on one reach correlate **0.78** under shared draws and -0.004 under independent ones, so a two-hundred-gauge book would look two hundred times better diversified than it is.

Resampling, not regeneration. The catalogue’s outcomes are already computed, so a simulated year is index lookups rather than a re-run of the hydrology. That is what makes the year count affordable.

Outputs, all from the same run:

- **Annual flood probability** — the occurrence view; what the PRS spread is priced from.
- **Expected floods per year** — the aggregate view. It exceeds the occurrence probability whenever a year can carry more than one flood, which is exactly the case a conditional-only model cannot represent at all.
- **Return-period levels** — quantiles of the annual distribution.
- **Event exceedance frequency** — valid below the one-year return period, which matters at frequent-flood gauges and cannot be expressed as a probability.

All outputs carry the frequency-model identifier and calibration provenance in their metadata.

5.9 The closed form, and why it is a test rather than the answer

For $N \sim \text{Poisson}(\lambda)$ with each event flooding independently with probability p :

$$\mathbb{E}[(1-p)^N] = \exp(\lambda((1-p) - 1)) = \exp(-\lambda p) \quad (4)$$

so $P(\text{at least one flood in a year}) = 1 - \exp(-\lambda p)$ **exactly**. The closed form is not an approximation of the simulation — it is the simulation’s expectation. The same argument gives the aggregate view directly as λp .

That is precisely what makes it valuable as a check: any gap beyond sampling error means one of the two is wrong. It is wired in as a reconciliation gate, not as the production pricing path, because the simulation additionally yields the whole annual distribution.

The gate measures the gap in **sampling standard errors**, $\sqrt{p(1-p)/n}$, against a configured band, rather than as a fixed percentage. A fixed band cannot mean the same thing at every year count: at ten thousand years a 2% band false-alarmed on 17% of runs, while at a million it never bound at all. The statistic is verified to be a true z -score — mean $|z|$ measures 0.811 and 0.797 at ten and a hundred thousand years against the theoretical $\sqrt{2/\pi} = 0.798$ — and that calibration is itself a test, so the band cannot silently stop meaning what it says.

5.10 Wiring

#	File	Change	State
1	src/models/hazard/io/_load.py	Sequence tagging; <code>event_id</code> and <code>count_events</code> ; <code>num_events</code> in metadata and summary	Built
2	src/models/hazard/builder.py:109--141	GEV fits per-event peaks; outputs renamed to <code>event_exceedance_prob</code> ; annual rates come from the simulation	Stage 3
3	src/port/src/property/hc/property.py:54,119,250	Property spread, gauge basis leg and return period become simulation-derived	Stage 4

5.11 Extension points

Architected now, built later: λ as a process rather than a constant (the interface takes an optional covariate or time argument); a per-peril λ registry so wind, fire and seismic attach without restructuring; and, because the simulation scores a boolean flood outcome per event, replacing that boolean with a loss quantum turns the same machinery into a full year loss table without changing the sampler.

6 The λ provenance problem

Finding. On synthetic catchments, calibrating λ from the gauge “historical” record is circular.

`src/models/statistics/synthetic.py:122--124` generates the daily series by injecting severe exceedances at exactly the rate implied by the gauge’s `FrequencyExceedLevel3` field:

```
1 severe_floods_expected = max(freq_exceed_level3, 1)
2 severe_flood_prob_per_day = severe_floods_expected / (years * 365)
```

and `FrequencyExceedLevel3` is itself produced by `src/port/rand/shared/gauge/gauge_field_generators.py:168--170` as a `random.choices(...)` draw over a five-year window.

A peaks-over-threshold extraction over that series therefore recovers the injected rate. **The only per-gauge frequency number the platform currently holds is a random integer.**

Consequences.

1. Version 1’s stated risk — “short or gappy records give a noisy $\hat{\lambda}$ ” — is the wrong risk. The real one is that λ has no observational content on synthetic catchments.
2. It does not invalidate the POT pipeline. POT is exactly what would be run against real gauge series, and building it now is correct.
3. It does change what Stage 1 may claim. The provenance class must record **generator-derived**, not **observed**, and the validation report must say so.

Approach adopted.

- **Now:** build POT as specified. On synthetic catchments, validate it by **round-trip** — the extracted $\hat{\lambda}$ must recover the injected rate within tolerance once declustering is accounted for. This is a genuinely strong test of the extraction code, and it is labelled as such rather than presented as calibration.

- **Follow-on** (separate approval; requires port regeneration): make `FrequencyExceedLevel3` a *modelled* field rather than a random one, giving the gauge generator a catchment-level λ prior so synthetic frequencies are physically plausible and spatially coherent. Gauges on the same reach should not carry independent λ .

7 Measured results

Everything in this section is measured, not estimated. It supersedes the corresponding estimates in version 2.0.

7.1 Cost and accuracy of the year simulation

Apple M2, 8 cores, 8 GB. A full portfolio is 2,200 subjects (gauges plus properties) scored against one shared set of draws.

Simulated years	Full portfolio	Sampling error
1,000	0.07 s	~5%
10,000 (<i>configured</i>)	0.40 s	~1.6%
100,000	4.1 s	~0.5%
1,000,000	43 s	~0.16%

The cost is linear in the year count and the error falls as its square root, so this is a knob that can be turned without touching code.

Run-to-run stability does not depend on the year count. The seed is pinned, so a re-run reprices identically at any setting. What the year count buys is closeness to the true expectation, not repeatability — a distinction worth holding onto, because the two are easily conflated when a number is described as “noisy”.

7.2 Repricing: withdrawn, not yet measurable

An earlier draft of this section reported a $5.40\times$ reprice (1301 bps to 7030 bps). That figure is withdrawn. It was not a model output. Two faults:

1. **The trigger level was invented.** The conditional came from an arbitrary threshold applied to a synthetic catalogue, not from a gauge’s severe-warning level. Nothing anchored it.
2. **The catalogue was resampled uniformly.** MKM-SS-001 generates its batches from 40% moderate, 35% severe, 20% extreme and 5% catastrophic — and *no* minimal or baseline at all — because it exists to train the stress classifier, where oversampling the interesting region is correct. Averaging over that uniformly answers $P(\text{flood} \mid \text{event is at least moderate})$, and multiplying *that* by a λ counting all qualifying events double-counts severity.

The tell was the implied return period: a severe flood every **0.81 years**. A conditional probability is easy to misread; a severe flood every nine months is obviously wrong. That check is now built in (`implied_return_period_years`) precisely so this class of error announces itself.

Fault 2 is fixed — the catalogue now carries population weights and the sampler draws proportionally, which moves the implied return period from roughly 1.9 to 3.4 years.

That is still not credible, and no repriced number should be quoted until it is. The remaining error is fault 1, and it cannot be fixed by choosing a better threshold by hand. It needs the constraint already present in the design:

$$\lambda_{\text{catchment}} \times P(\text{flood} \mid \text{event}, g) \equiv \text{gauge } g\text{'s measured severe-exceedance rate} \quad (5)$$

Until that identity is asserted against real gauge trigger levels, any conditional this layer produces is unfalsifiable. **Anchoring it is Stage 3 work, and no repricing figure belongs in this document before then.**

Two open questions the anchoring will settle, recorded so they are not lost:

- The population weights are engineering-judgement seeds. How often a qualifying Thames storm event is severe-or-worse is the input that most directly sets the answer, and it wants an owner's view rather than a modeller's guess.
- Whether $\lambda = 4.5/\text{yr}$ and the seeded weights are *jointly* consistent with observed severe-flood return periods, or whether one of them has to move.

7.3 Extraction against the existing calculation

On a thirty-year synthetic gauge record, at the severe trigger:

Measure	Rate
Exceedance days per year — what the statistics module reports today	1.234
Declustered events per year	0.967

A **28% overstatement**, being the mean flood duration in days. This is the quantity the fix-and-promote removes.

The decomposition identity also holds exactly on that record: $\lambda \times P(\text{severe} \mid \text{event}) = 0.9672$ against a directly measured severe rate of 0.9672. That identity is the per-gauge validation arm.

8 Non-functional requirements

Restated against the project coding rules.

Rule	Requirement for this project
R1 — no parameters outside <code>config</code>	All thresholds, declustering windows, selection criteria, fallback λ and return-period grids live in <code>config/frequency/</code> . The orphan <code>storms_per_year=20</code> is absorbed.
R2 — no file over 300 lines	Package layout adopted from the outset.
R3 — $\geq 99\%$ coverage, per stage	Verified at the end of <i>each</i> stage. Version 1 specified 100% branch coverage; aligning to the house rule avoids a bespoke standard for one package. Plus property-based tests for mean and variance recovery, and a golden-master calibration regression.

Rule	Requirement for this project
R4 — no functions in <code>__init__</code> <code>--py</code>	Re-exports only.
R5 — canonical copyright header	On every new source file; enforced by the self-healing header audit.
R6 — data store only via database	Zero direct SQL, ORM or connections; zero JSON artefacts.

Additionally: seeded RNG throughout, with calibration and simulation reproducible from (data version, configuration version, seed); a structured run log carrying inputs, configuration hash, code version, outputs and validation flags; and verification through the full audit — *not* `split.audit.py`, which version 1 cited and which no longer exists. Relevant full-audit sections: 4.3 path definitions, 4.5 JSON files, 4.6 database usage, 4.7 model chain consistency.

9 Project plan

Leaves first; each stage independently shippable and tested; coverage checked at every stage boundary.

The staging has changed from version 2.0. The year sampler was version 2.0's Stage 6 and is now built, because it became the engine rather than a deferred extension. Stage 1 is correspondingly larger and partly done.

Stage	Content	State	Reprices?
1a	Event-definition alignment; event catalogue; peaks-over-threshold extraction with declustering; per-catchment λ ; <code>num_events</code>	Done	Denominator only, not yet consumed
1b	Monte Carlo year sampler; shared draws; closed-form reconciliation gate	Done (<i>was Stage 6</i>)	Not yet consumed
1c	Day-count fix-and-promote; λ and provenance persisted through the <code>database</code> seam	To do	No
2	Poisson and NegBin families; dispersion selection; override logging; round-trip validation	To do	No
3	Simulation wired into <code>builder.py</code> ; legacy metric behind a deprecation flag; parallel-run repricing report on the real book	To do	Yes
4	Property and gauge legs in <code>_process.py</code> ; return periods; clamp review; wind-leg decision	To do	Yes
5	Governance: inventory entry, model documentation, registry wiring, auto-generated validation report, monitoring job	To do	No

Stage	Content	State	Reprices?
6	<i>Separate approval:</i> loss-weighted year loss table, ELT export, wind λ calibration, seasonal rates	To do	—

9.1 Stage acceptance

- **S1a/S1b** — *met.* Event-granular loader covered; catalogue and sampler under test; the simulation reconciles with its closed form; no pricing path consumes λ . 204 tests, 100% coverage on the frequency packages and the hazard I/O module.
- **S1c** — peaks-over-threshold series reproducibly generated for all gauges; provenance complete and persisted through the seam; the day-count consumers repointed.
- **S2** — families pass property tests for mean and variance recovery; calibration deterministic under fixed inputs; per-gauge selection report generated; round-trip recovers the injected rate within tolerance.
- **S3** — parallel-run report produced on the real book, quantifying repricing per gauge; legacy metric still available behind the flag; simulated and closed-form annual probabilities reconcile within the sampling band.
- **S4** — property-level repricing quantified; clamp and wind-leg decisions recorded; user interface and end-of-day consumers verified against the renamed fields.
- **S5** — validation report reviewed and signed off; monitoring scheduled; legacy metric formally deprecated.

10 Migration landmines

These bite between Stage 3 and Stage 5. Each needs a decision before Stage 3 lands.

#	Landmine	Detail	Status
L1	Magnitude of the reprice	$\lambda \equiv 1$ today, so a reprice is certain and material. Its size is not yet measurable — the earlier $5.40\times$ figure is withdrawn (Section 7) and no replacement should be quoted until the conditional is anchored against real gauge trigger levels.	Open, size unknown. The Stage 3 parallel-run report on the real book is the gating business artefact. No switchover without it, and no number before it.
L8	Population weights are load-bearing and unvalidated	The conditional scales directly with the seeded event-population weights. A wrong severe-or-worse share moves every spread proportionally, and nothing currently contradicts a wrong value.	Open. Needs an owner's view on how often a qualifying Thames event is severe-or-worse, and the Stage 3 anchoring to make a wrong value fail loudly.

#	Landmine	Detail	Status
L2	Compensating hacks become distortions	<code>builder.py:87</code> clamps the exceedance probability up to 0.01; <code>MAX_RETURN_PERIOD</code> : 100 caps return periods. Both exist <i>because</i> return periods were fake.	Once return periods are real, both must be reviewed or removed. Removing them widens the tail — quantify alongside L1.
L3	<code>num_storms</code> is persisted and rendered	A stored field read by the user interface, the blotter and end-of-day.	Resolved. <code>num_storms</code> kept, <code>num_events</code> added beside it. Verified against the CDM property editor, hazard, storm-gauge and intensity suites — no consumer changed.
L4	End-of-day history spans the cut	Stored <code>annual_hazard_rate*</code> values change semantics <i>and</i> magnitude; historical series become non-comparable across the switchover date.	Mark the cut in the end-of-day record; do not silently backfill.
L5	Per-storm flood events collapse	Under an event definition, several storms in a sequence become one event; per-property flood counts drop even before λ is applied.	Open at the property leg. Measured at the gauge leg: 2.34 storms per event, and the conditional <i>rose</i> rather than fell. The property pipeline still counts per storm.
L6	Wind leg diverges	The wind union also divides by <code>num_storms</code> . If flood is annualised and wind is not, the BOW and BAW union and joint legs become internally inconsistent.	Open. Either move both at Stage 4 or explicitly freeze the wind leg on the legacy metric with a recorded justification.
L7	Correlation is now load-bearing	Portfolio risk depends on subjects sharing one set of event draws. A future caller that ran the single-subject convenience wrapper per subject would silently decorrelate the book — measured at 0.78 versus -0.004 .	Open. Guard at the portfolio entry point when Stage 4 wires it, and treat as a review point for any new caller.

11 Governance and registration

Per the new-model checklist, and noting that a new model document must be wired into **five registries** plus the Makefile filter:

1. Inventory entry MKM-EF-001 — tier 1 (proposed), category Hazard, source module `src/models/frequency/`, upstream MKM-GH-001, downstream MKM-PR-001.
2. Corresponding upstream and downstream amendments on MKM-GH-001 and MKM-PR-001; the chain is hand-maintained and scanned by full-audit section 4.7.
3. `docs/models/event_frequency/event_frequency.tex` and Makefile, with the standard sections and the generated test-results and sensitivity inputs.
4. Registry wiring: five registries plus the Makefile filter.
5. Assumptions and limitations recorded — including, explicitly, the provenance limitation of Section 6 and the stationarity assumption.

Validation plan (SR 11-7). Conceptual soundness; per-gauge count backtesting on a held-out split; dispersion coverage reporting (share of gauges selecting NegBin); threshold sensitivity ($\hat{\lambda}$ and AAL under a one-band threshold perturbation); reconciliation of analytical against empirical AAL; the POT round-trip; annual recalibration with a drift alert; and an effective-challenge artefact emitted on every run.

Note that benchmarking return-period levels against published national flood-frequency estimates — version 1’s validation test 6 — is **not meaningful** while catchments are synthetic. It is retained in the plan but flagged as pending real gauge data.

12 Risks and assumptions

Risk or assumption	Impact	Mitigation
λ has no observational content on synthetic catchments	Pricing is precise but not accurate; validation claims could overstate	Provenance class recorded on every fitted rate; round-trip framed as extraction validation, not calibration; real-data path built and ready
Repricing of 2–4× on every quote	Client and book impact	Stage 3 parallel-run report before switchover; staged rollout; legacy metric retained behind a flag
Event-definition change reprices before λ does	Two effects confounded in the impact analysis	Sequenced deliberately: Stage 1 isolates the denominator effect, Stage 3 the λ effect
Threshold choice materially moves $\hat{\lambda}$	Pricing instability	Sensitivity test; threshold governance in configuration with a change log

Risk or assumption	Impact	Mitigation
Overdispersion ignored where present	Understated tail	NegBin selected wherever the dispersion test rejects Poisson
Stationarity assumed	Understated forward risk under climate change	Explicit limitation in the model documentation; drift monitoring; Track B roadmap
Wind leg left on the legacy metric	Internally inconsistent peril comparison	Decide explicitly at Stage 4; do not leave implicit
Clamp removal widens the tail	Compounds the repricing impact	Quantify jointly with the repricing report, not separately

13 Glossary

AAL

Average Annual Loss; expected loss per year.

AEP

Aggregate Exceedance Probability; the probability that total annual loss exceeds x . *Track B*.

OEP

Occurrence Exceedance Probability; the probability that the largest single event loss in a year exceeds x . *Track B*.

EEF

Event Exceedance Frequency; the annual frequency, not probability, of events exceeding a level.

ELT / YLT

Event Loss Table and Year Loss Table. *Track B*.

Event

Here, the insurance hours-clause event — a storm *sequence* within the configured event window — not an individual storm.

POT

Peaks Over Threshold; extreme-value sampling of all independent exceedances above a threshold.

GPD

Generalised Pareto Distribution; the standard model for peaks-over-threshold exceedance magnitudes.

Dispersion index

Variance divided by mean of annual counts; near one supports Poisson, above one indicates overdispersion.

NHPP

Non-Homogeneous Poisson Process; a time-varying arrival rate. *Track B*.

Provenance class

observed, generator-derived or regional-fallback; records what a fitted λ is actually based on.